# Problem set 6 Solution
# Logistic Regression

## Excercise 1

1. Let $\overline{W}^T = (w_1, w_2, \ldots, w_d)$ and $\overline{P}^T = (p_1, p_2, \ldots, p_d)$ be points from $R^d$ and let $b$ be a number from $R$. Compute the distance from $\overline{P}$ to the hyperplane defined by the equation $\overline{W}^T\overline{X} + b = 0$.

## Solution 1

We use the fact that the vector $\overline{W}$ is perpendicular to the hyperplane $\overline{W}^T\overline{X}+b = 0$.

Let $L$ be the line via the point $\overline{P}$ and parallel to the vector $\overline{W}$, that is $L = \{\overline{P} + t\overline{W} \mid t \in R\} \subseteq R^d$. Line $L$ intersects the hyperplane $\overline{W}^T\overline{X} + b = 0$ when $\overline{W}^T(\overline{P} + t\overline{W}) + b = 0$, i.e. when

$$t = \frac{-(\overline{W}^T\overline{P} + b)}{||\overline{W}||^2}.$$

The distance between this point of intersection and the starting point $\overline{P}$ is

$$d = ||\overline{P} + t\overline{W} - \overline{P}|| = ||t\overline{W}|| = |t| \cdot ||\overline{W}|| = \frac{|\overline{W}^T\overline{P} + b|}{||\overline{W}||^2} \cdot ||\overline{W}|| = \frac{|\overline{W}^T\overline{P} + b|}{||\overline{W}||}.$$

The latter is proportional to $|\overline{W}^T\overline{P} + b|$.

## Excercise 2

Show the following properties of the logistic sigmoid function $\sigma(x) = \frac{1}{1+e^{-x}}$.

1. $\sigma(x) = 1 - \sigma(-x)$
2. $\frac{\partial\sigma}{\partial x} = e^{-x}\sigma^2(x)$

## Solution

1. $1 - \sigma(-x) = 1 - \frac{1}{1+e^x} = \frac{1+e^x-1}{1+e^x} = \frac{e^x}{1+e^x} \cdot \frac{e^{-x}}{e^{-x}} = \frac{1}{e^{-x}+1} = \sigma(x)$
2. $\frac{\partial \sigma}{\partial x} = \frac{e^{-x}}{(1+e^{-x})^2} = e^{-x}\sigma^2(x)$

## Excercise 3

### Solution

The odds of an event are defined as

$$\text{Odds} = \frac{p}{1-p}$$

where $p$ is the probability of the event occurring.

**1. Odds when** $p = \frac{1}{2}$

$$\text{Odds} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$$

Thus, the odds are 1, which is often written as $1:1$. This means that the event is just as likely to occur as it is not to occur.

**2. Log odds**

The log odds (also called the *logit*) are defined as

$$\log\left(\frac{p}{1-p}\right)$$

Substituting $p = \frac{1}{2}$:

$$\log\left(\frac{\frac{1}{2}}{1-\frac{1}{2}}\right) = \log\left(\frac{\frac{1}{2}}{\frac{1}{2}}\right) = \log(1) = 0$$

Thus, the log odds are 0.

**3. Interpretation**

When the probability of an event is $\frac{1}{2}$:

- The odds are 1, meaning the event and the non-event are equally likely.
- The log odds are 0, which represents a neutral point.

More generally:

- If $\log\left(\frac{p}{1-p}\right) > 0$, then $p > 0.5$ and the event is more likely to occur.
- If $\log\left(\frac{p}{1-p}\right) < 0$, then $p < 0.5$ and the event is less likely to occur.

Therefore, a log odds value of 0 corresponds to a probability of 0.5, indicating equal likelihood of the event occurring or not occurring.

## Exercise 4

Suppose we collect data for a group of students in a statistics class with variables $X_1$ = hours studied, $X_2$ = undergrad GPA, and $Y$ = receive an A. We fit a logistic regression and produce estimated coefficient, $\beta_0 = -6$, $\beta_1 = 0.05$, $\beta_2 = 1$.

1. Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.
2. How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

**Solution**

The logistic regression model is given by

$$P(Y = 1 \mid X_1, X_2) = \frac{e^{\beta_0+\beta_1 X_1+\beta_2 X_2}}{1 + e^{\beta_0+\beta_1 X_1+\beta_2 X_2}}$$

where the log-odds form is

$$\log\left(\frac{P(Y = 1 \mid X_1, X_2)}{1 - P(Y = 1 \mid X_1, X_2)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Given the estimated coefficients

$$\beta_0 = -6, \qquad \beta_1 = 0.05, \qquad \beta_2 = 1$$

the model becomes

$$P(Y = 1 \mid X_1, X_2) = \frac{e^{-6+0.05X_1+X_2}}{1 + e^{-6+0.05X_1+X_2}}$$

1. **Probability of getting an A**

   For a student who studies 40 hours and has GPA 3.5,

   $$X_1 = 40, \qquad X_2 = 3.5$$

   First compute the linear predictor:

   $$z = -6 + 0.05(40) + 1(3.5)$$

   $$z = -6 + 2 + 3.5 = -0.5$$

   Now compute the probability using the logistic function:

   $$P(Y = 1) = \frac{e^{-0.5}}{1 + e^{-0.5}}$$

$$e^{-0.5} \approx 0.6065$$

$$P(Y=1) = \frac{0.6065}{1+0.6065}$$

$$P(Y=1) \approx 0.3775$$

$$\boxed{P(Y=1) \approx 0.38}$$

Thus, the estimated probability that the student receives an A is approximately 0.38 (or 37.8%).

2. **Hours needed for a 50% chance of getting an A**

   A 50% probability occurs when

   $$P(Y=1) = 0.5$$

   For logistic regression, this happens when the log-odds equals zero:

   $$-6 + 0.05X_1 + 1(3.5) = 0$$

   Simplify:

   $$-6 + 3.5 + 0.05X_1 = 0$$

   $$-2.5 + 0.05X_1 = 0$$

   $$0.05X_1 = 2.5$$

   $$X_1 = \frac{2.5}{0.05} = 50$$

   $$\boxed{X_1 = 50}$$

   Thus, the student would need to study **50 hours** to have a 50% chance of receiving an A.